BI10 - ROUND 1 - DECODE

Prepared by

Nguyễn Trần Thảo Nhi (Team Leader)

Nguyễn Tuấn Kiệt

Trần Nguyễn Thu Hà

Trương Vũ Phúc

September 2026

EXECUTIVE SUMMARY

Business Context: ITB, a Vietnamese consumer-finance company, sought to move from a reactive, collections-oriented view of customers toward a proactive financial-wellbeing view - understanding how customers spend, how healthy their finances look month to month, and how engaged they are with digital channels. This analysis uses a synthetic 2025 dataset (1,852,394 transactions; 999 customers; 34 provinces) to build customer segments and design non-punitive support actions. This is not a credit-risk scoring exercise.

Key Findings Across Five Analytical Tasks

Task 1 - Spending Behavior: Annual spending rose 179.89% from February to December (peak: VND 488.0B), driven by transaction volume (+187.33%) rather than larger purchases. Financially stressed customers (health < 40) allocate 71.6% of spend to discretionary categories vs. 40.5% for Healthy customers (⩾ 80). Digital adoption in high-spending provinces Hanoi (40.88%) and Đồng Nai (40.73%) lags the national average (41.18%). In-store Groceries leads total spending (VND 513.77B), while Fuel & Transportation leads transaction frequency (188,029). Gen Z (< 25) transacts most (221.6/month) with low essential spending (35.49%); Young Professionals (25-34) show the lowest resilience (65.79) and highest spend-to-income ratio (71.58%).

Task 2 - Financial Health: 72.46% of customers are Stable (60-79), only 4.32% reach Healthy (⩾ 80), 22.35% are At-Risk, and 0.86% are severely stressed. Spend-to-income ratio (r = -0.90) and credit utilization (r = -0.86) are the strongest negative health drivers. A Stressed but Highly Engaged segment (59 customers, 5.91%) transacts 131 times/month while spending 199% of income - a debt spiral needing support, not credit restriction.

Task 3 - Customer Engagement: 73.99% of consumer-months are Highly Engaged, 25.57% Moderately, and 0.44% Disengaged. Despite this, 58.54% of spend still flows through offline POS vs. 41.47% through digital channels. Recency and frequency differentiate engagement most strongly (Disengaged: 19.7 inactive days vs. near-zero for engaged groups). A relative threshold (bottom 20% engagement among Healthy customers) identifies a practical target of 75 customers (7.51%), since a strict absolute definition yields only 1.

Task 4 - Customer Segmentation: K-Means clustering (k = 4, silhouette = 0.312) covers 100% of the 999 - customer base: Healthy & Highly Engaged (49.4%), Stretched but Highly Engaged (41.6%), Emerging Digital (8.4%), New & Early-Dormant (0.5%). The Stretched segment holds 54 of the 59 Danger Zone customers identified in Task 2.

Task 5 - Business Recommendations: Six non-punitive interventions - budgeting tools, spend alerts, financial-planning reminders, financial education, digital-channel nudges, and product suggestions - reach 62 to 250 customers each, each grounded in a specific data-driven finding.

Credit Policy Statement: financial_health_score is a wellbeing indicator only. It is never used to approve, deny, reduce, or cancel credit access for any customer, at any point in this analysis or its recommendations.

TABLE OF CONTENTS

01 **Introduction to the Case**

02 **Task 1 - Exploratory Data Analysis**
Spending trends, essential vs. discretionary allocation, regional digital adoption, category behavior, age-cohort vulnerability

03 **Task 2 - Financial Health Analysis**
Score distribution, key drivers of low health, demographic/geographic variation, the "Stressed but Highly Engaged" segment

04 **Task 3 - Customer Engagement Analysis**
Engagement distribution, channel usage, category diversity, recency & frequency, the "Healthy but Under-Engaged" segment

05 **Task 4 - Customer Segmentation**
Feature engineering, K-Means clustering methodology, four customer segments, validation against a rule-based benchmark

06 **Task 5 - Business Recommendations**
Six non-punitive interventions, target segments, reach, and prioritization

07 **Limitations & Future Work**

INTRODUCTION TO THE CASE

Business Background

ITB is a Vietnamese consumer-finance company that issues cards and short-term credit products to retail customers. Leadership wants to understand customer financial wellbeing, spending behavior, and digital engagement in order to design proactive, supportive interventions - rather than make credit-risk decisions. This case is explicitly NOT a credit-scoring exercise: financial_health_score must never be used to approve, deny, reduce, or cancel credit access.

Dataset Overview

Two linked synthetic datasets cover calendar year 2025, joined by consumer_id:

- consumer_transactions_2025 - 1,852,394 transaction-level records covering 999 customers, 693 merchants, and 34 provinces/cities. Contains category, merchant, channel, payment method, and time - of - day detail.
- consumer_financial_health_engagement_2025 - 10,992 consumer-month records (999 customers × up to 12 months). Contains financial_health_score, engagement_score, and all ratio-based indicators (essential/discretionary spend ratio, online spend ratio, spend-to-income ratio, credit utilization ratio, spending volatility, category diversity, recency).

All monetary fields (income, credit limit, balances) are synthetic and inflated relative to real Vietnamese salaries due to the density of the underlying simulation - all cross-customer comparisons in this analysis therefore rely on ratio-based metrics, never raw VND values.

Data Quality

- 1,852,394 transactions verified with no duplicate transaction_id and no missing critical fields.
- All timestamps fall within calendar year 2025; each consumer_id maps to a single stable demographic profile.
- Customers with fewer than 12 months of history (new/mid-year joiners) are flagged and treated as provisional in aggregate calculations.
- Ages range from 14 to 96 (includes minors, retained per the dataset's simulated supplementary/student-card design — noted, not filtered).

Methodology

Analysis uses Python (pandas, seaborn, scikit-learn) on Google Colab. Task 1 - 3 operate at the consumer-month grain (10,992 rows); Task 4 - 5 aggregate to the customer grain (999 unique customers), using sum for monetary totals, mean for ratios and scores, and the most recent month for recency - computed only over months each customer was active.

MONTHLY SPENDING ANALYSIS

VND 3.24T

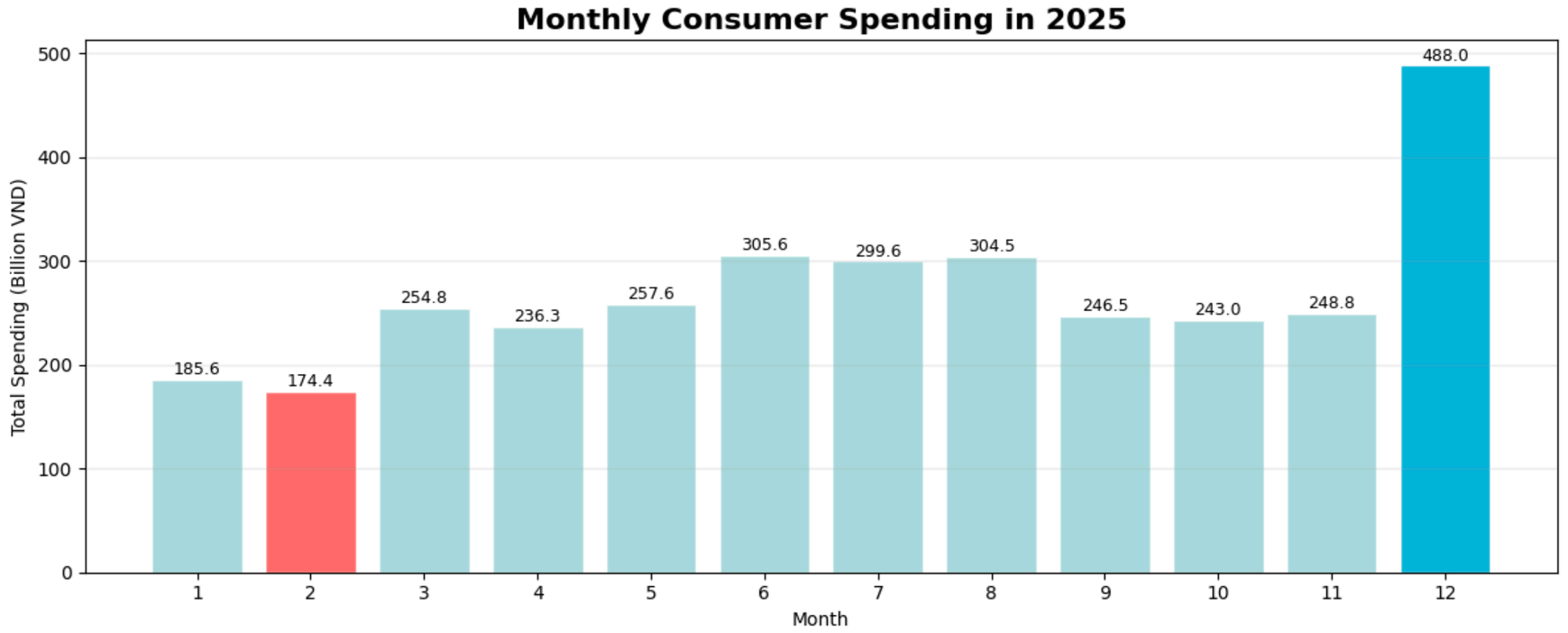
Total Annual Spending

+179.89%

Spending Growth (Feb–Dec)

+187.33%

Transaction Volume Growth



Transaction Volume Drives Spending Growth

Consumer spending increased significantly throughout 2025, reaching its highest level in December.

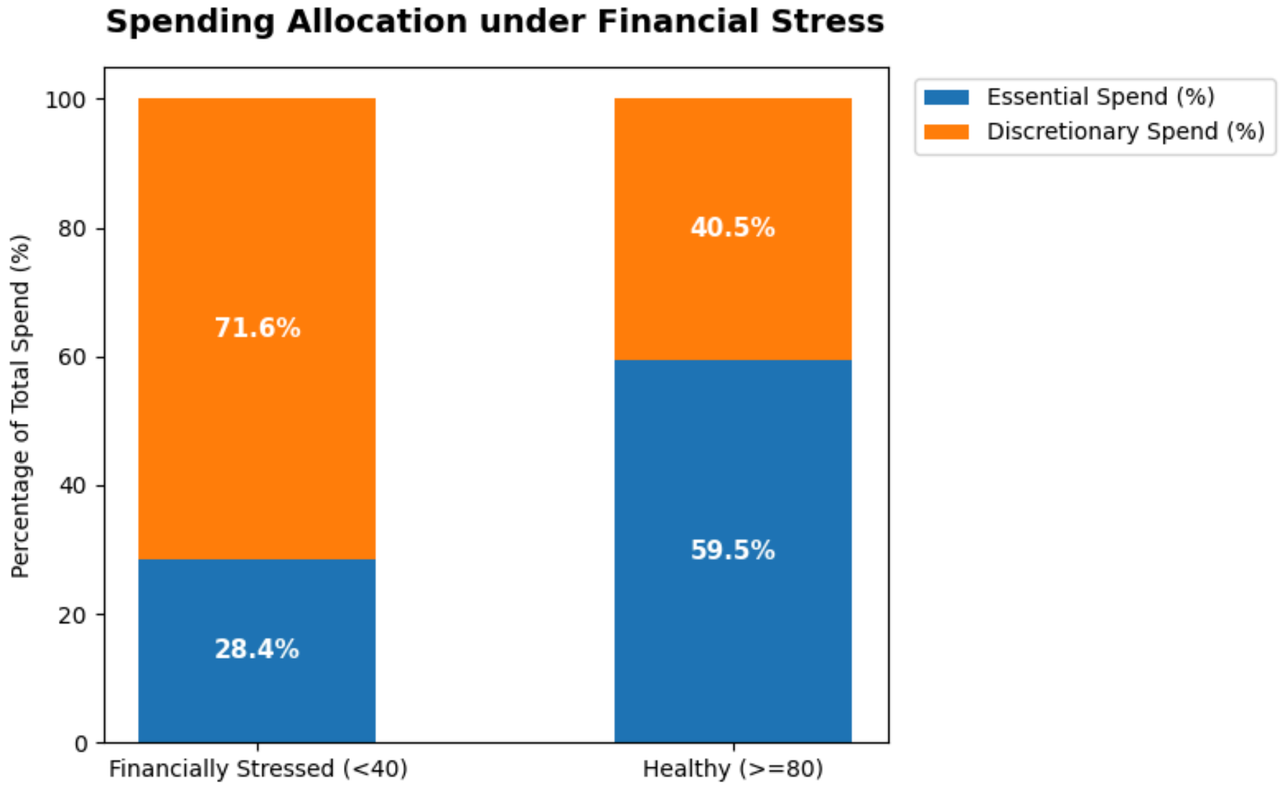
The increase was primarily driven by transaction frequency rather than larger individual purchases.

SPENDING ALLOCATION UNDER FINANCIAL STRESS

71.6%
Discretionary Allocation — Stressed (<40)

59.5%
Essential Allocation — Healthy (⩾80)

1.77x
Stressed vs. Healthy Discretionary Ratio



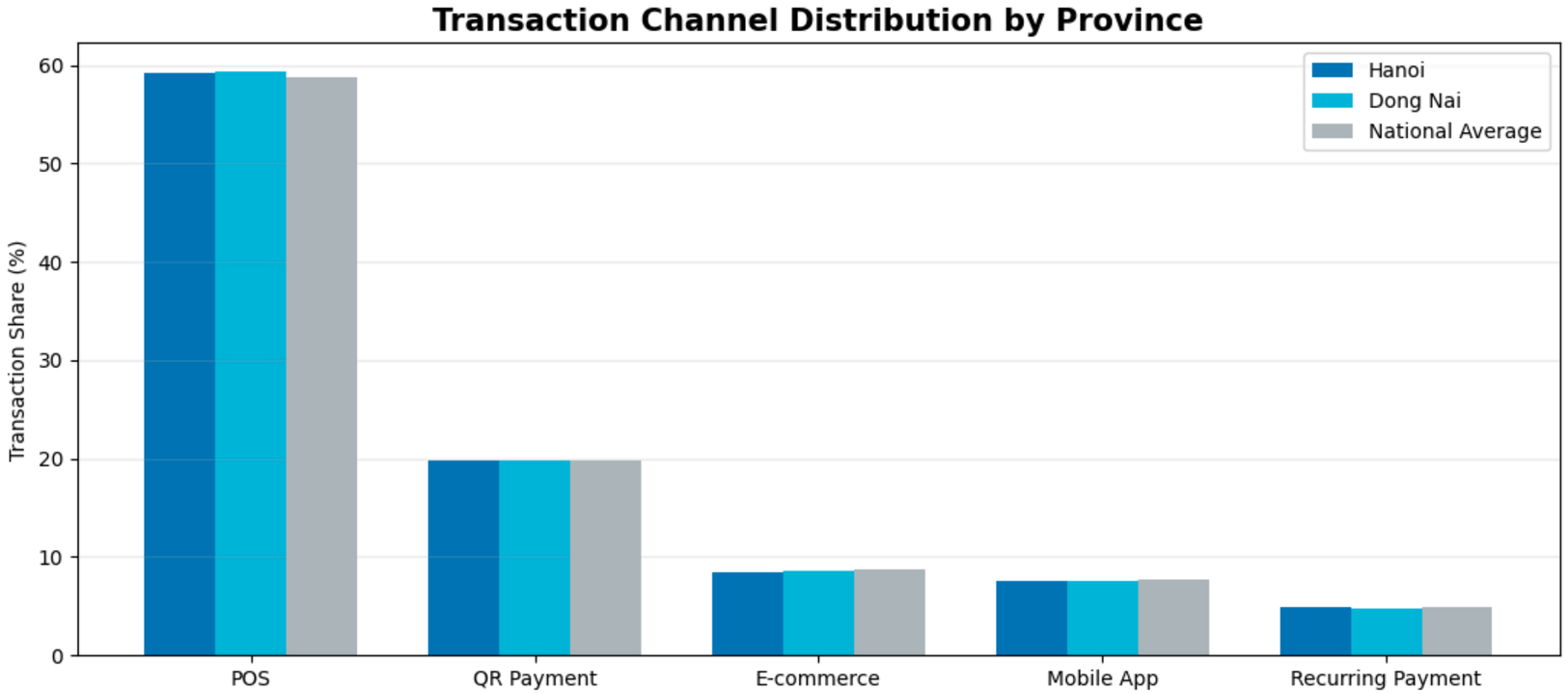
The Discretionary Spending Trap Drives Financial Distress

- Reversed Allocation: Stressed users (<40) sink **71.6**% of spend (VND 24.20B) into non-essentials, leaving only **28.4**% for essentials.
- Healthy Benchmark: Healthy users (⩾80) prioritize resilience, allocating **59.5**% (VND **33.17B**) to essentials and capping discretionary spend at **40.5%**.
- Root Cause: Distress stems from lifestyle inflation and unconstrained non-essential spending, not rising basic living costs.

Strategic Takeaway: Deploy automated category spending caps and smart nudges when discretionary spend exceeds 50% of monthly income.

TRANSACTION CHANNEL DISTRIBUTION

41.18%	40.88%	40.73%
National Digital Share	Hanoi Digital Share	Dong Nai Digital Share



Digital Adoption Gaps in High-Spending Provinces

Hanoi and Dong Nai rank 2nd and 3rd in total spending, reaching VND 267.26B and VND 183.07B, respectively.

However, their digital transaction shares remain below the national average of 41.18%:

- Hanoi: 40.88% (−0.31 pp)
- Dong Nai: 40.73% (−0.46 pp)

These gaps suggest opportunities to encourage digital payment adoption in high-spending markets.

Digital channels = QR Payment + E-commerce + Mobile App + Recurring Payment (excluding POS).
Source: BI10 Consumer Transaction Dataset (2025), Task 1 – Q3.

CONSUMER SPENDING CATEGORIES

VND 513.77B

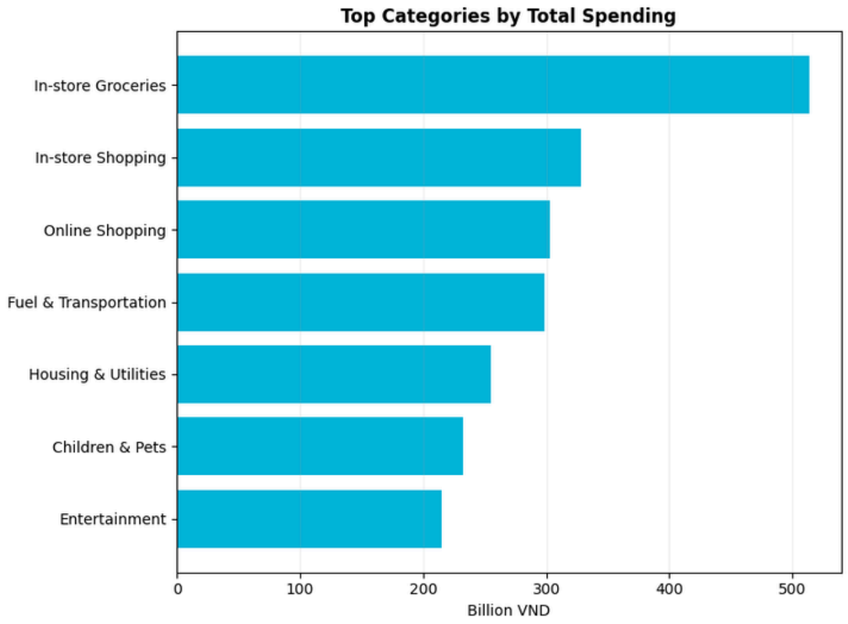
Highest Spending – In-store Groceries

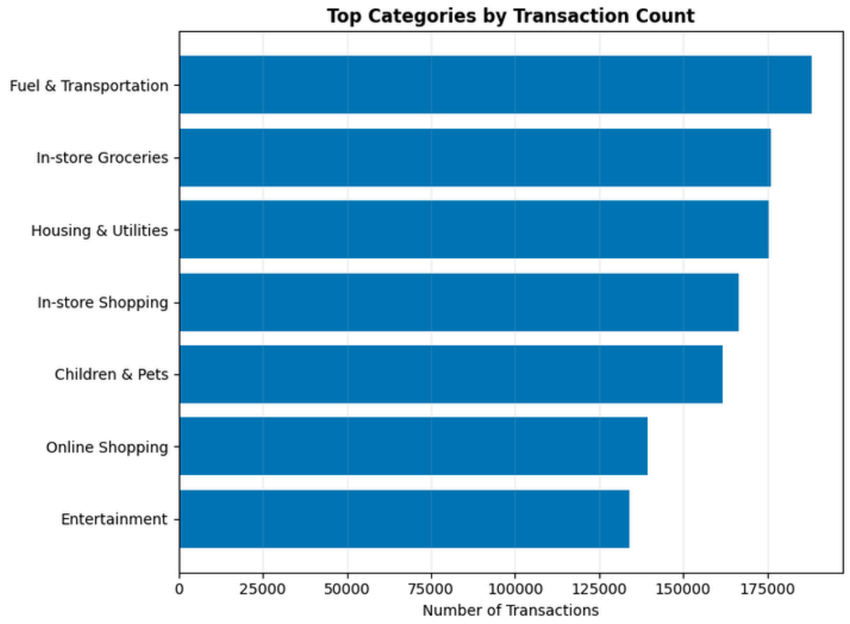
188,029

Most Transactions – Fuel & Transportation

1.84×

Groceries vs. Transportation Average Ticket





Spending Value and Frequency Tell Different Stories

In-store Groceries leads total spending (VND 513.77B; 15.83% of annual spending), with an average ticket of VND 2.92M.

Fuel & Transportation records the highest transaction frequency (188,029; 10.15% of transactions), with a lower average ticket of VND 1.59M.

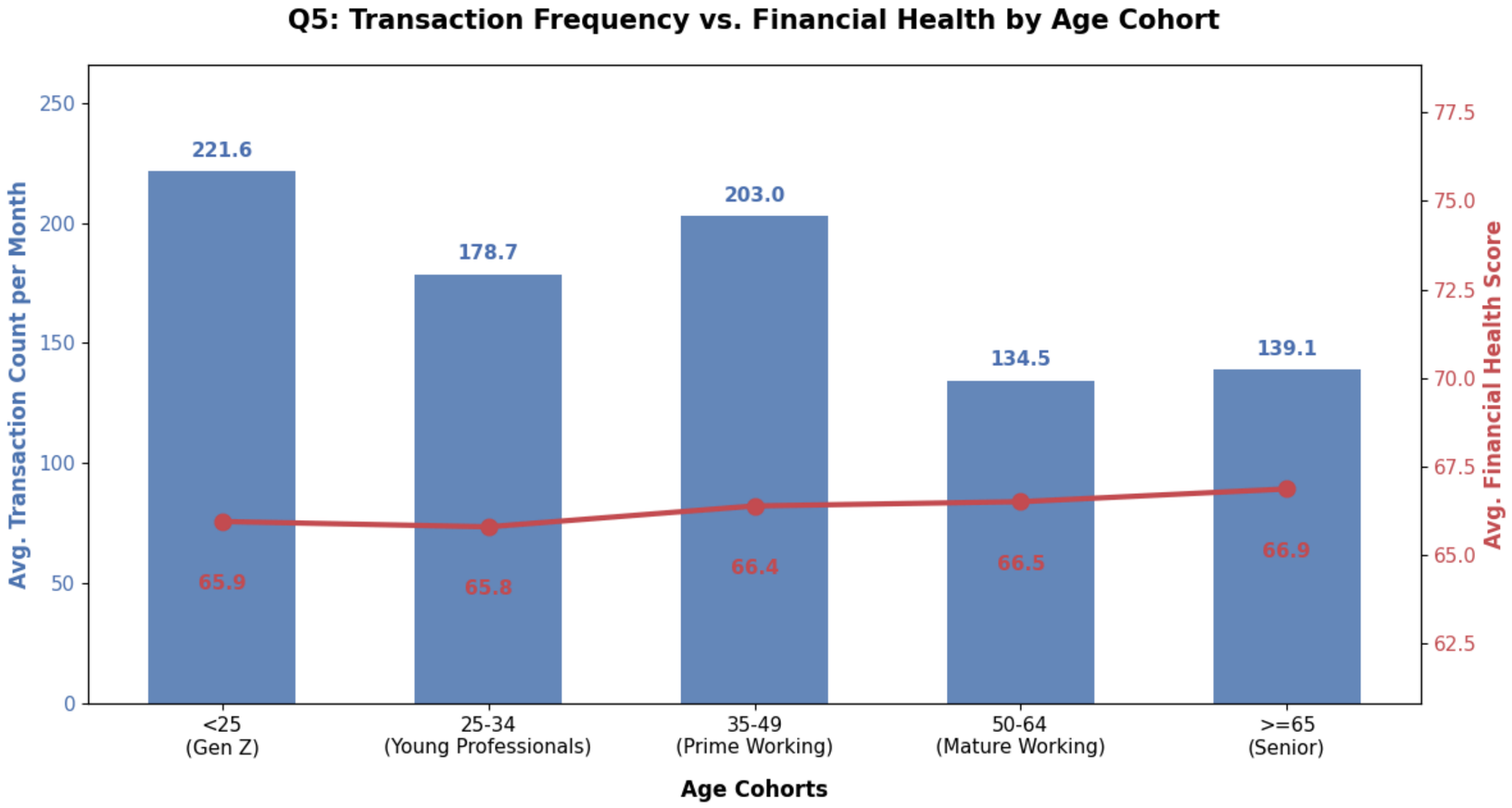
This highlights the difference between high-value purchases and frequent, lower-value transactions.

AGE COHORTS & FINANCIAL RESILIENCE

221.6
Highest Monthly Tx — Gen Z (<25)

65.79%
Lowest Health Score — 25–34 Cohort)

64.51%
Discretionary Spend Share — Gen Z (<25)



Younger Cohorts Drive Velocity but Face Financial Fragility

- Gen Z (<**25**) Activity Paradox: Records the peak transaction frequency (**221.6** tx/month) alongside an average health score of **65.94**. Essential spending accounts for only **35.49%**, indicating high exposure to discretionary purchases.
- Young Professionals (**25–34) Squeeze**: Experiences the lowest resilience score across cohorts **(65.79)**, compressed by the highest spend-to-income ratio (71.58%) and a credit utilization rate of **19.49%**.
- Mature Stability (**35**+): Health scores steadily recover with age (**66.38 to 66.86)** as essential allocations stabilize above 50%.
- *Strategic Takeaway*: Roll out automated micro-saving tools and discretionary spend alerts targeted specifically at users under 35.

TASK 2. FINANCIAL HEALTH ANALYSIS
CONSUMER SPENDING CATEGORIES

Healthy Gap

4.32%

Gap in wealth products
(>80 Score, 475 Records)

Stable Majority

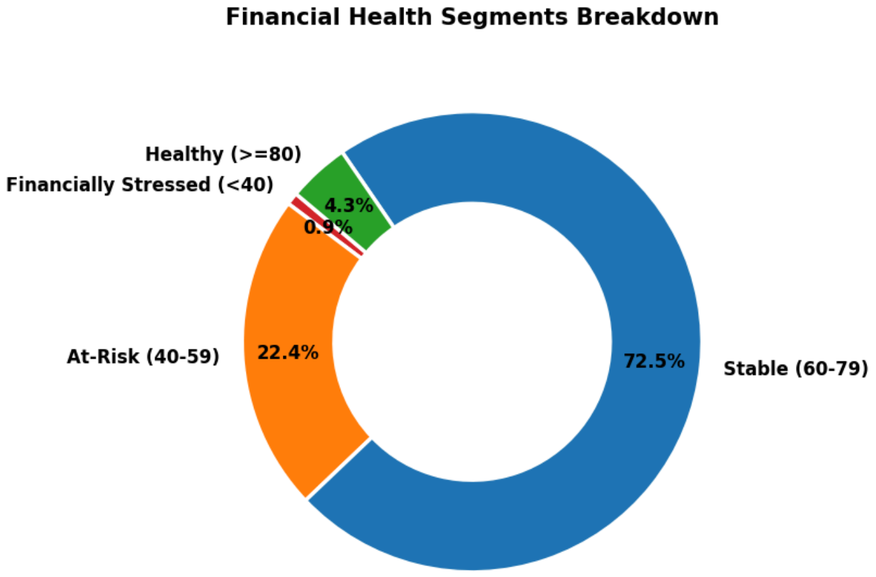
4.32%

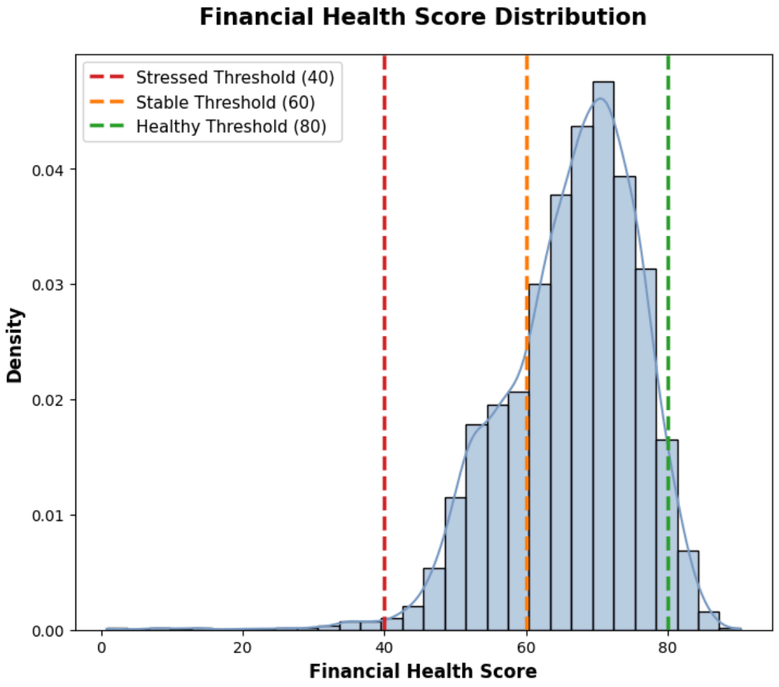
High credit quality
(60-79 Score, 7965 Records)

At - Risk Warning

22.35%

Susceptible to macroeconomic risk
(40-59 Score, 2457 Records)





Stable Foundation Masks Latent Vulnerabilities

While **72.46%** of consumers sit securely in the Stable tier (60–79), reflecting solid baseline credit quality, the portfolio faces a dual challenge:

- Uncaptured Upside: Only **4.32%** reach the **Healthy threshold (⩾80)**, leaving substantial headroom for wealth and savings cross-sell opportunities.
- Downside Exposure: Over **22.35%** are concentrated in the At-Risk zone **(40–59)**. Although severe distress (<40) remains low at **0.86%,** this sizable At-Risk cohort represents a major default risk buffer if exposed to macroeconomic shocks.

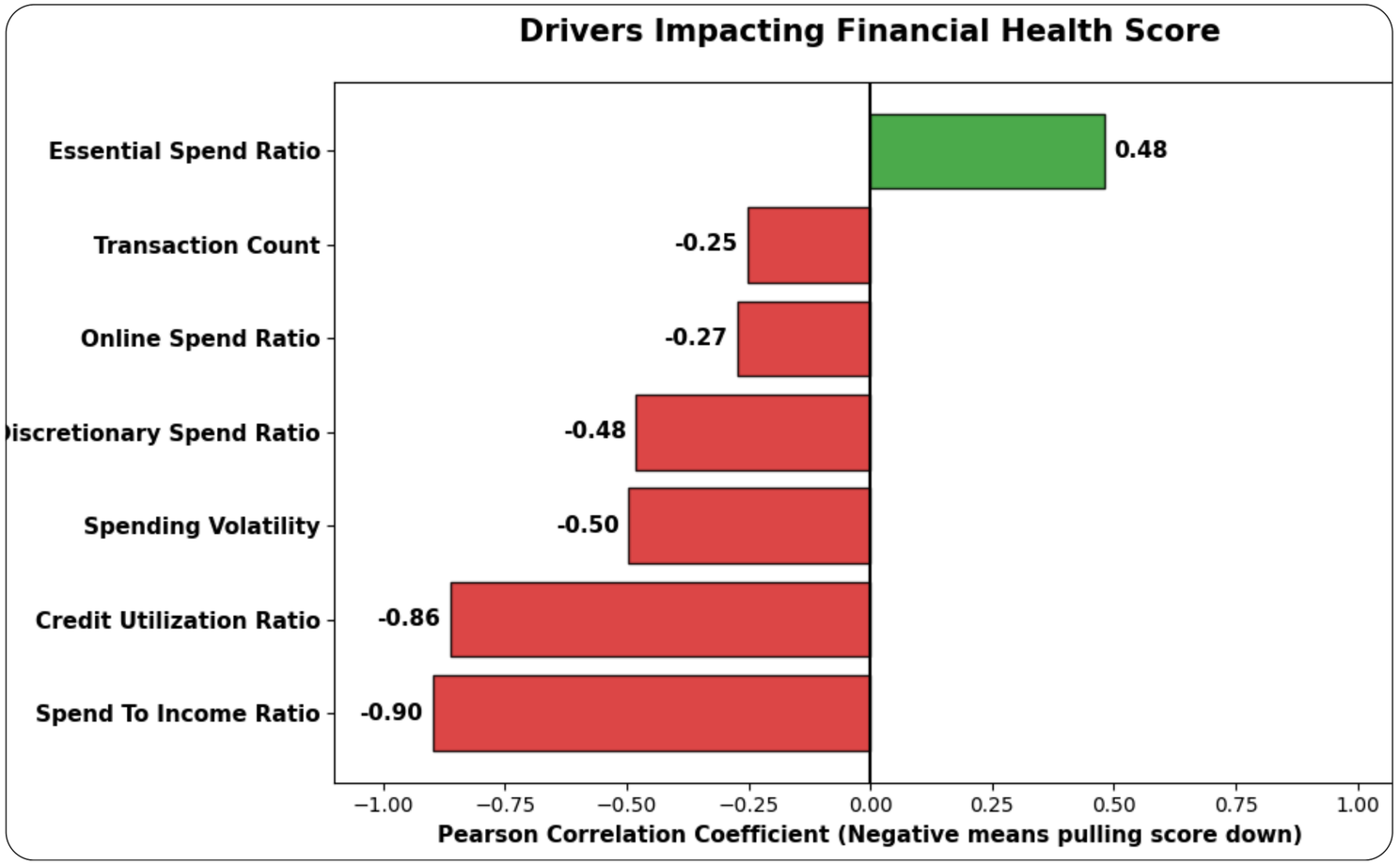
Strategic Takeaway: The priority is shifting focus from reactive debt recovery toward **preemptive risk management** for the At-Risk base and up-tiering Stable users into wealth products.

THE "DESTROYERS" OF FINANCIAL HEALTH

-0.90
Top Risk Driver
(Spend-to-Income Ratio)

-0.86
Leverage Pressure
(Credit Utilization Ratio)

+ 0.48
Protective Shield
(40-59 Score, 2457 Records)



Core Drivers (Strong Negative Correlation)

- The two primary culprits tanking health scores are living beyond one's means (Spend to Income Ratio, r = -0.90) and maxing out credit limits (Credit Utilization Ratio, r = -0.86), both being 'default risk' indicators.

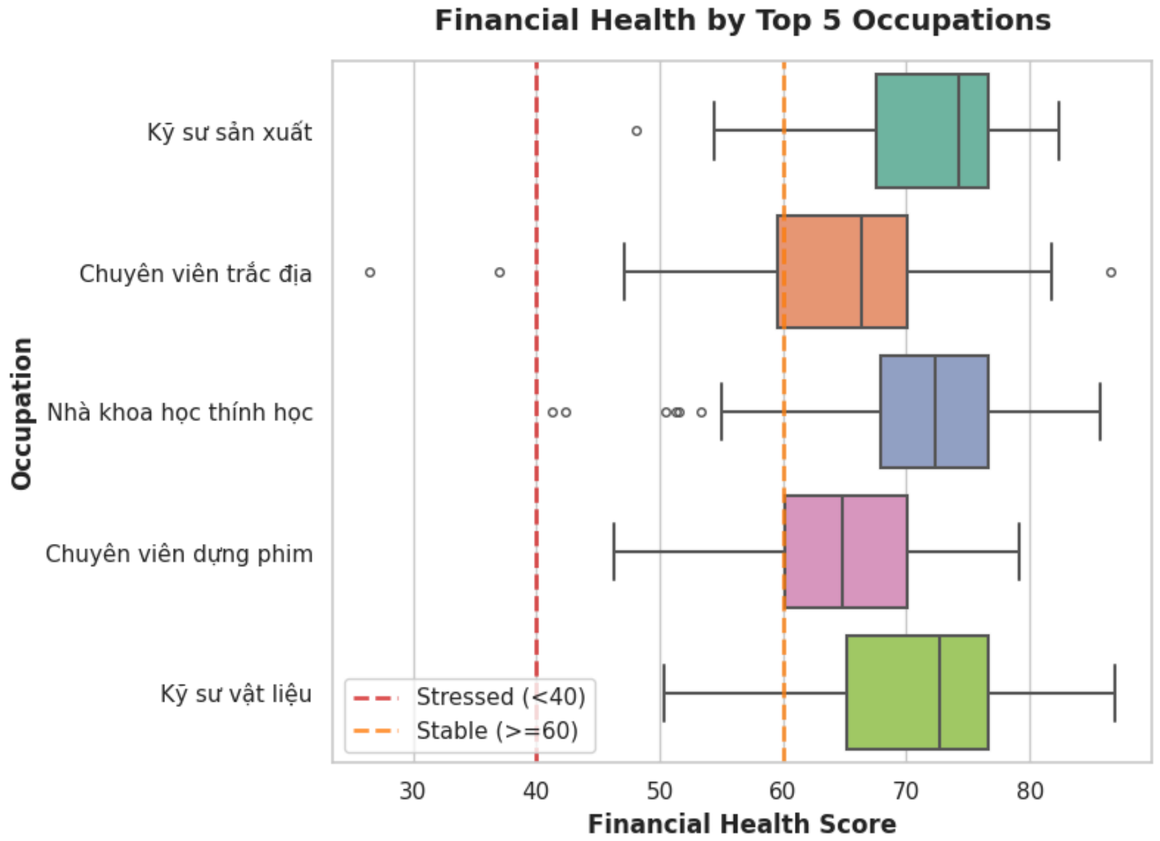
Behavioral Accelerators (Moderate Negative Correlation)

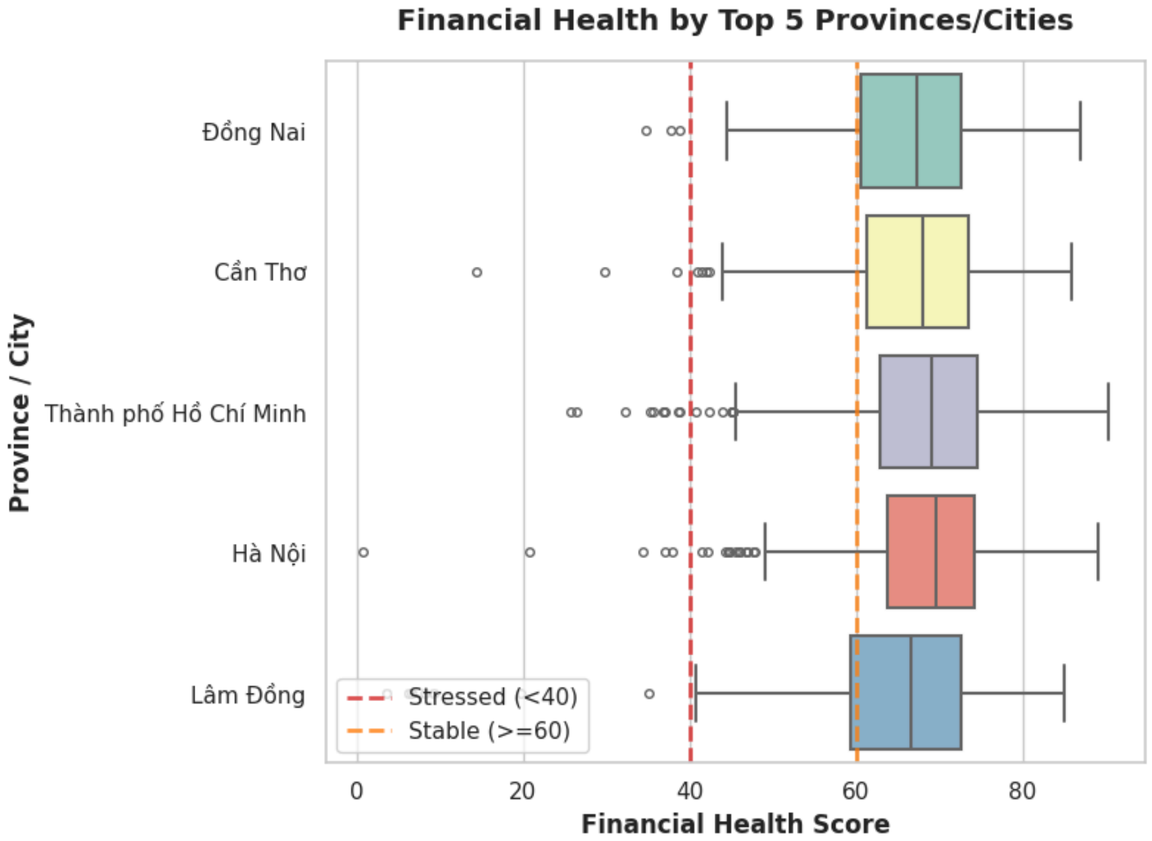
- Spending Volatility (r = -0.50) and Discretionary Spend Ratio (r = -0.48) are catalysts depleting buffers. Boomss, as impulsive spending leads to fragility.

Protective Shield & Strategic Implication

- Essential Spend Ratio (r = +0.48) is the defensive buffer. ITB must deploy preemptive Utilization Alerts (at 30% and 50% thresholds) to trigger automated Installment Loans for debt restructuring before default.

DEMOGRAPHIC & GEOGRAPHIC RISK DIVERGENCE





- **Occupational Divergence:** Engineering roles anchor portfolio resilience with strong medians (74.2 & 72.6) and zero exposure below 48.0, whereas Surveyors exhibit severe downward vulnerability with outliers plunging to 26.5 (<40 Stressed tier). Rather than relying on uniform occupational limits, ITB should institute dynamic credit buffers and preemptive check-ins tailored to career cash-flow volatility.

- **Urban Risk Blindspots**: While regional medians appear uniformly sound across all five hubs (66.5–69.6), headline averages mask severe localized tail-risk: major metropolitan centers conceal extreme cash-flow depletion, plunging to lows of 0.8 in Hanoi and 25.7 in HCMC. ITB must look beyond regional aggregates to deploy hyper-local living-cost alerts and proactive liquidity safeguards for metro cardholders.

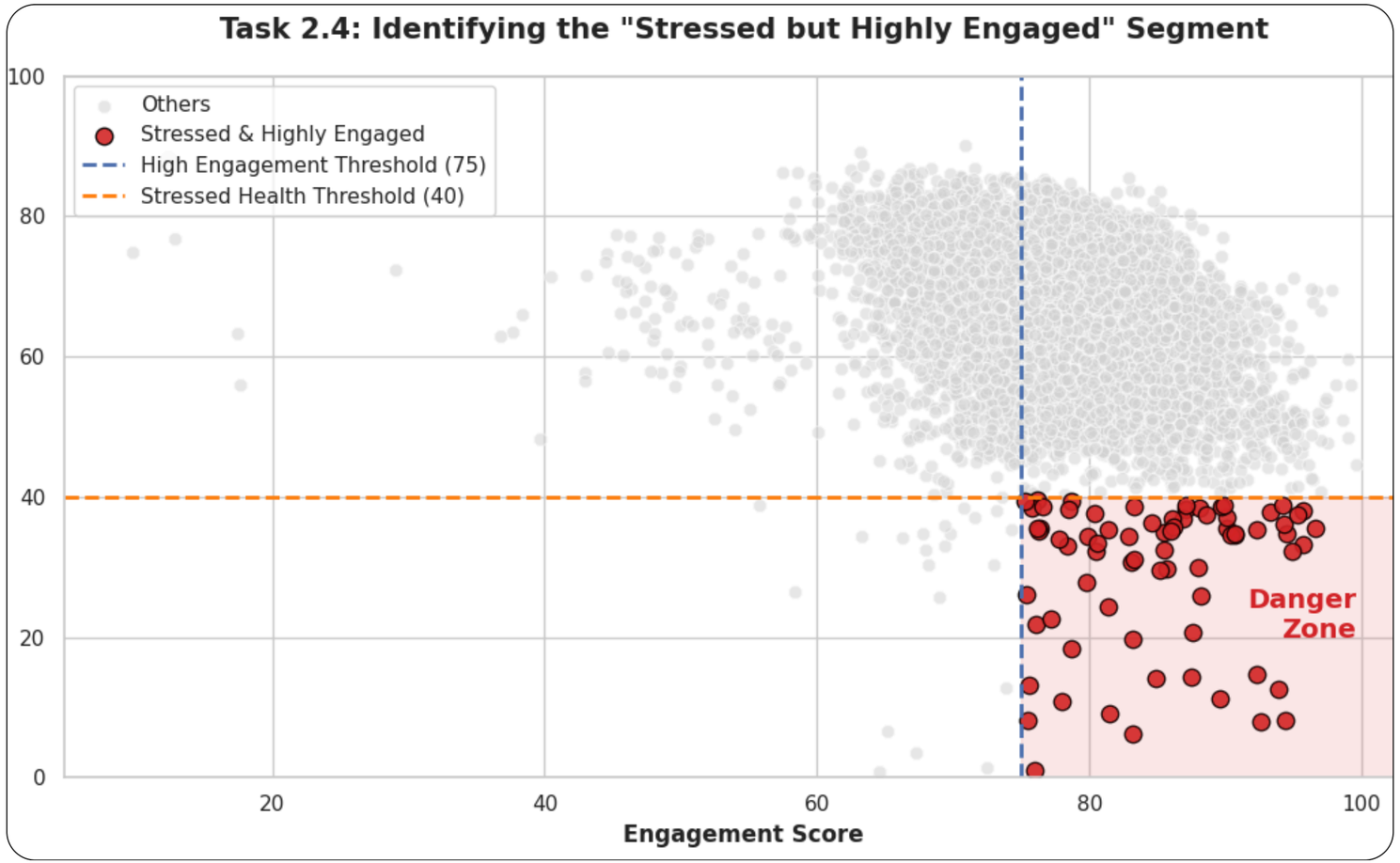
Based on Top 5 Occupations (n = 505) and Top 5 Provinces (n = 3,817)

THE "DANGER ZONE": HIGH ENGAGEMENT VS. FINANCIAL STRESS

5.91%
Cohort Size
(59 Target Customers)

199%
Average Spend-to-Income Ratio

131
Avg. Monthly Transactions



The High-Activity Debt Spiral

- Segment Size & Boundary: Defined by a strict, reproducible filter (**Health Score < 40 & Engagement Score ⩾ 75**), this cohort comprises 59 customers (**5.91**%) concentrated in the lower-right quadrant.
- Behavioral Paradox: Despite critical distress, these users transact aggressively (**131** tx/month) while burning **199**% of monthly income, relying heavily on revolving credit lines to support unsustainable spending.
- Non-Punitive Intervention: Abrupt card cancellations risk inducing immediate systemic default. Instead, ITB should deploy automated debt restructuring into fixed-rate installment loans, combined with discretionary spending alerts within the mobile banking app.

TASK 3 - CUSTOMER ENGAGEMENT ANALYSIS

CUSTOMER ENGAGEMENT DISTRIBUTION & CHANNEL USAGE

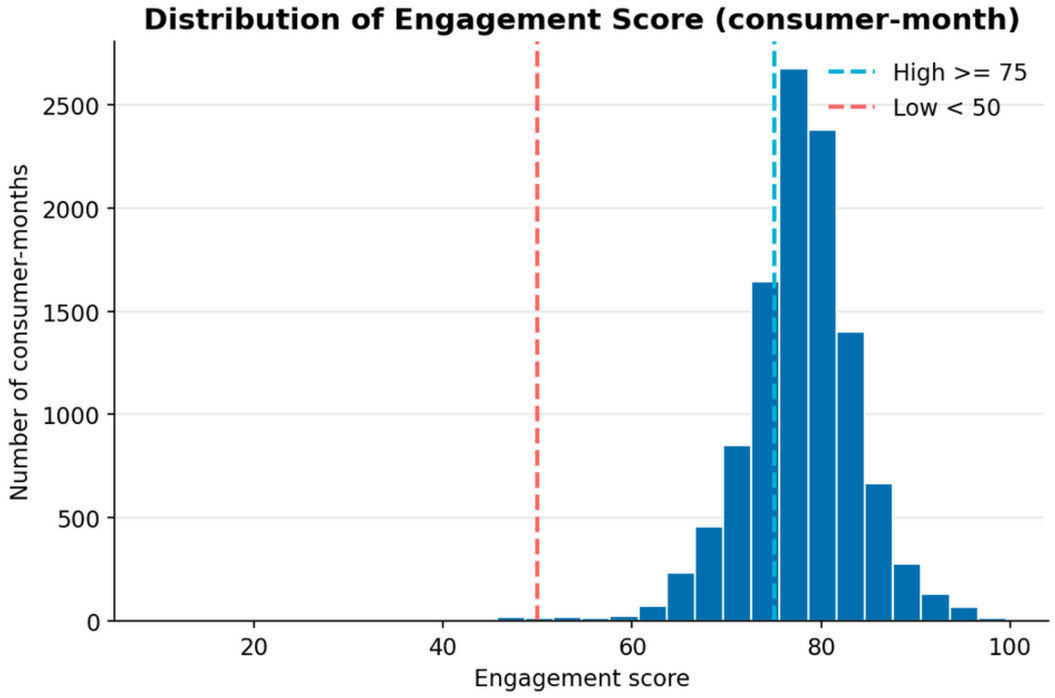
Highly Engaged Customers
73.99%
(n = 8,133)

Moderately Engaged
25.57%
(n = 2,811)

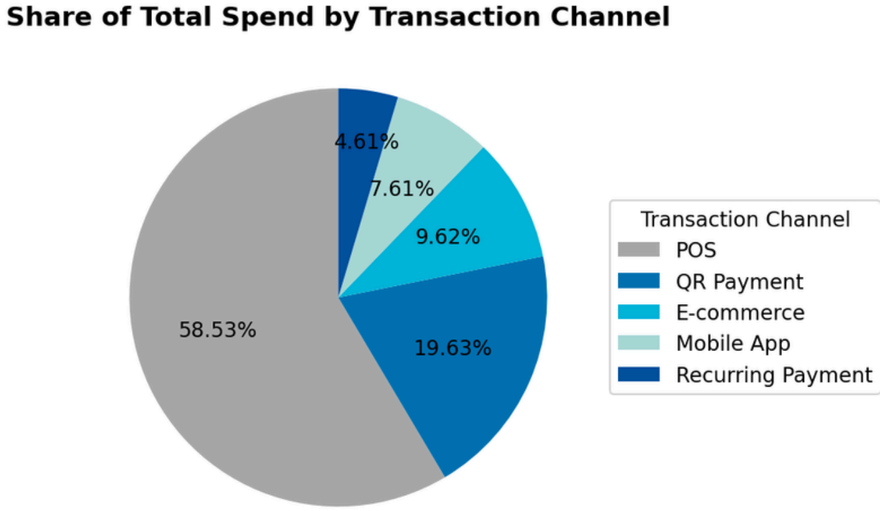
Disengaged
0.44%
(n = 48)

Spend via POS
58.54%
(Offline)

Spend via digital channels
41.47%
(Online)



Note: Disengaged customers represent only 0.44% (n = 48) of all consumer-months - a negligible share relative to Highly and Moderately Engaged segments.



Segment	Definition
Highly Engaged	Engagement score ⩾ 75
Moderately Engaged	Engagement score 50 -75
Disengaged	Engagement score < 50

The engagement score distribution across **10,992 consumer-months** is heavily right-skewed: the majority of the mass sits between 70-90, well above the "Highly Engaged" cutoff of 75.

Business takeaway:
Despite very high overall engagement, channel adoption still skews traditional - over half of spend still flows through physical POS.

CATEGORY DIVERSITY

More Categories Used

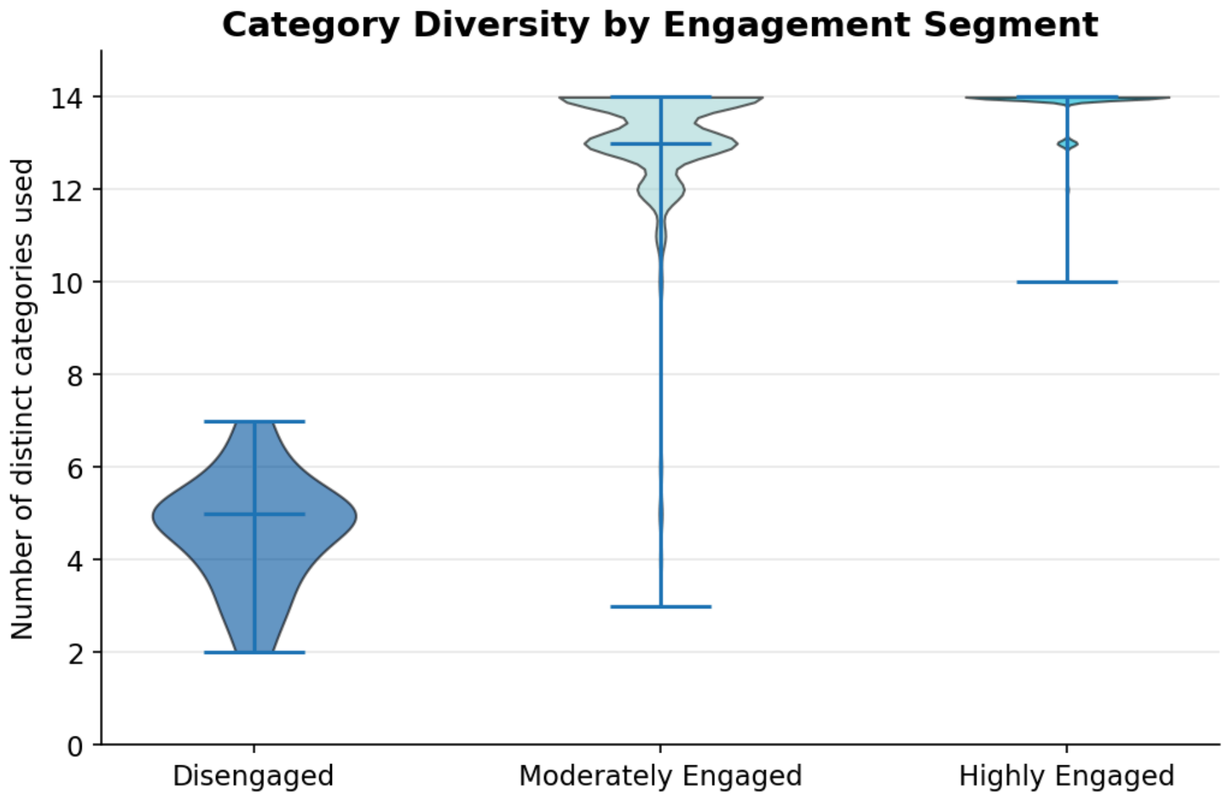
3x

(Highly vs. Disengaged)

Median Category Diversity

14

(Overall Customer Base)



◇ marks a statistical outlier within the Highly Engaged group's distribution.

Category diversity (number of distinct spending categories used) ranges from 2 to 14 categories, with an overall median of 14 - indicating that most customers already use nearly the full category range, another sign of the dataset's high baseline activity.

The violin plot shows the Disengaged group tightly clustered around 4 - 5 categories, while the Moderately Engaged group shows a bimodal shape (one cluster near 10 - 11, another near 14) suggesting this "middle" segment is not fully homogeneous and could be split further in future work.

Mean category diversity by segment

- Disengaged: 4.62
- Moderately Engaged: 13.17
- Highly Engaged: 13.89

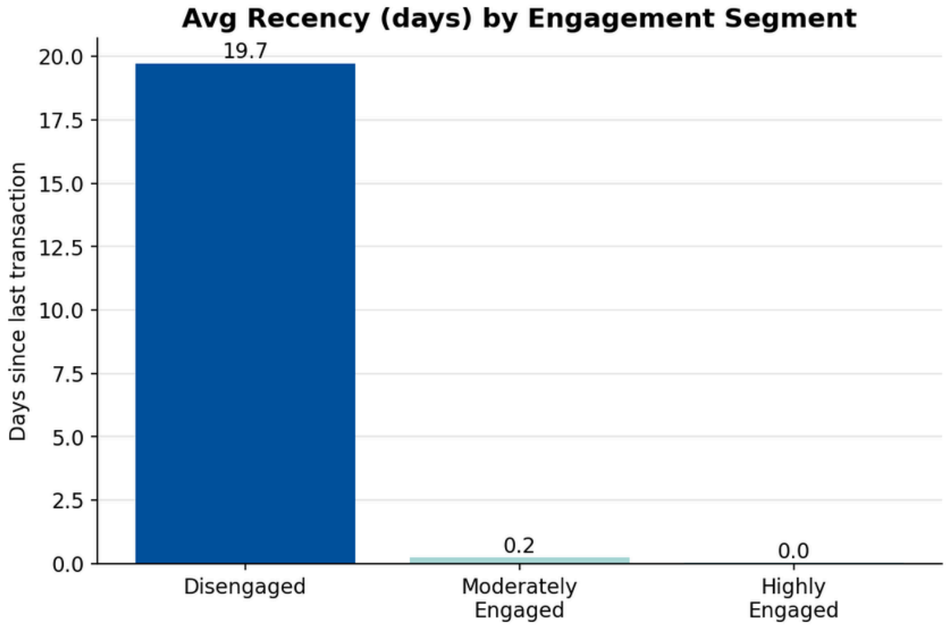
Business takeaway: Category diversity clearly separates truly disengaged customers from active ones, but is less effective at distinguishing degrees of engagement among already-active customers. The bimodal pattern within Moderately Engaged suggests two distinct sub-behaviors - worth revisiting as a potential refinement in Task 4 segmentation.

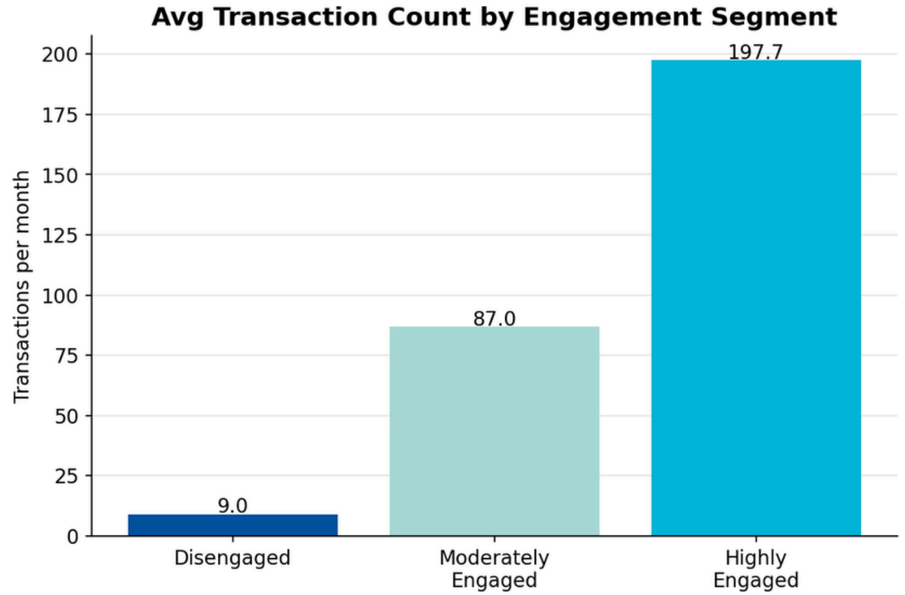
RECENCY & FREQUENCY VS ENGAGEMENT

More Transactions/Month
~22x
(Highly vs. Disengaged)

Median Recency
0 Days
(Overall Customer Base)

Avg. Inactivity
19.7 Days
(Disengaged Segment)





Recency (days since last transaction) ranges from 0 to 30 days (**median = 0**)

Recency is the strongest single differentiator of engagement: Disengaged customers average **19.7 days** since their last transaction, versus **0.2 days** for Moderately Engaged and just **0.0 days** for Highly Engaged.

Transaction frequency (transactions per month) ranges from 2 to 713 **(median = 150)**

Transaction frequency shows the same pattern:

- Disengaged: 9.0/month
- Moderately Engaged: 87.0/month
- Highly Engaged: 197.7/month.

Business takeaway: recency and frequency move almost in lockstep and together identify the Disengaged group with very high confidence - this makes recency the most practical early-warning signal for churn monitoring, more actionable than category diversity alone.

HEALTHY BUT UNDER - ENGAGED: THRESHOLD DEFINITION

Under Strict Threshold

n = 1

(Health⩾80 & Engagement<50)

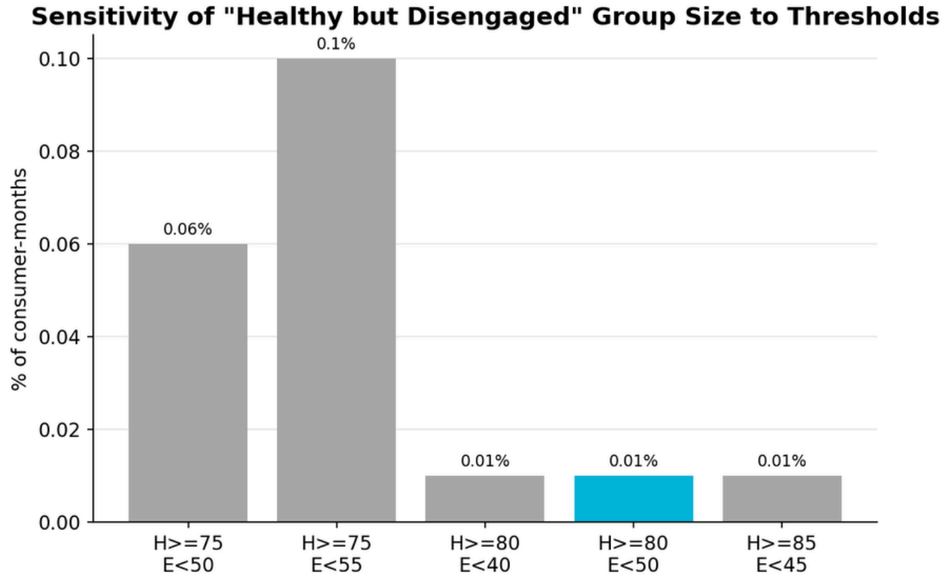
Under Relative Threshold

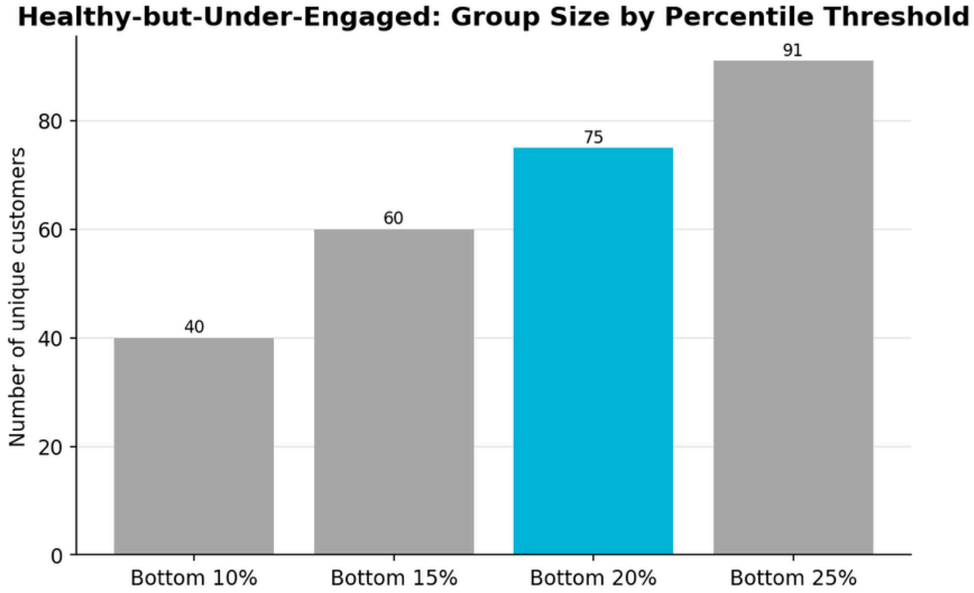
75 Customers

(Bottom 20% Percentile)

Share of Total Customer Base

7.51%





Only **n = 1 consumer-month (0.01%)**, corresponding to **1 unique customer.**

All return n between 1 and 11 rows - This confirms the pattern is a structural feature of the dataset rather than an artifact of the chosen cutoff - engagement is skewed high across the entire customer base, consistent with the very high baseline transaction activity observed throughout this analysis.

Interpretation: under a strict absolute definition, financially healthy-yet-fully-disengaged customers are essentially non-existent in this dataset.

Among the 475 consumer-months with **financial_health_score ⩾ 80**, we ranked engagement scores and tested the bottom 10/15/20/25th percentiles.

Chosen definition: Bottom 20% of engagement scores among Healthy customers (engagement_score ⩽ 68.0) → n = 97 consumer-months, 75 unique customers = 7.51% of the total 999-customer base.

HEALTHY BUT UNDER - ENGAGED: TARGET SEGMENT PROFILE

Under Relative Threshold

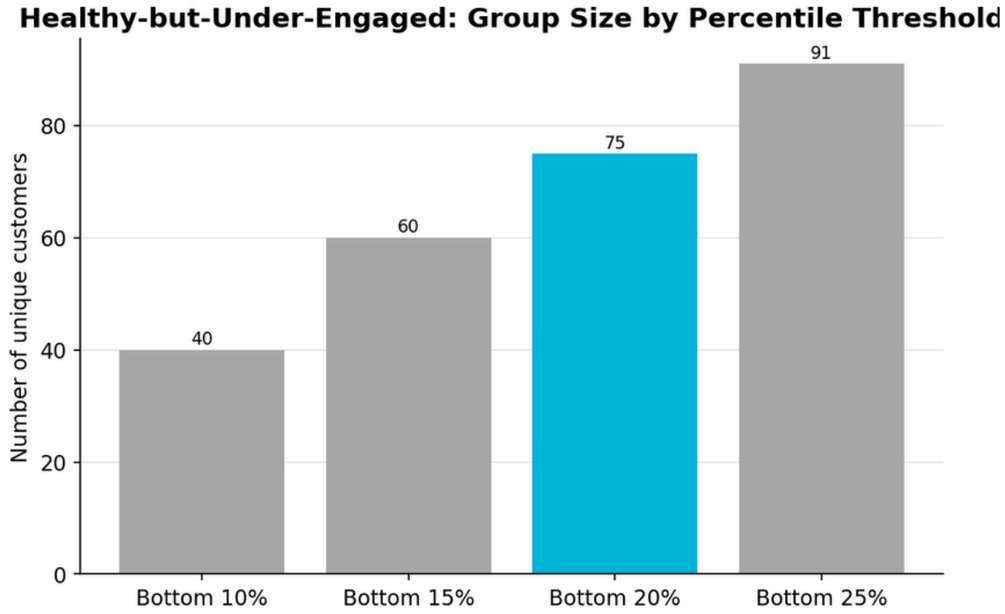
75 Customers

(Bottom 20% Percentile)

Share of Total Customer Base

7.51%

(75 / 999)

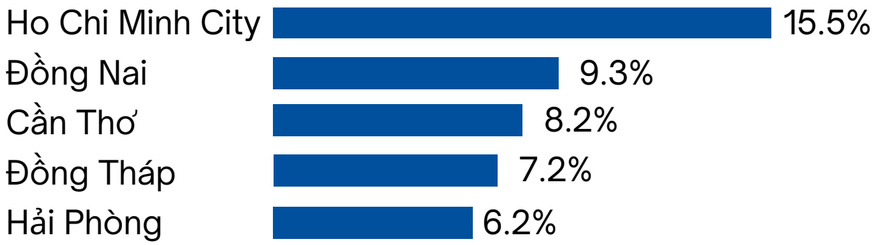


Why Bottom 20%?

- 10% (40 customers): reach too small for a meaningful intervention
- 25% (91 customers): dilutes the "under-engaged" distinction
- 20% (75 customers): balances actionable reach with a meaningful gap vs. peers within the Healthy segment

Note: the engagement cutoff (68.0) falls within the "Moderately Engaged" band, not below 50 - so this group should be labeled "Healthy but Relatively Under-Engaged", not "Disengaged," to avoid overstating the finding.

Provincial concentration among this segment:



Note: percentages calculated over 97 consumer-months (not unique customers); a customer appearing in multiple months may be counted more than once.

Demographic profile of this group (n=97): occupation is highly dispersed (top occupation "Kỹ sư vật liệu / Materials Engineer" at only 6.2%) - occupation is not a useful targeting variable here. Province shows more concentration (Ho Chi Minh City 15.5%, Đồng Nai 9.3%, Cần Thơ 8.2%) - suggesting a mild geographic pattern worth flagging but not over-interpreting given small per-province samples.

Business takeaway: this segment (~75 customers) is the practical target for retention-focused, non-punitive engagement nudges in Task 5 - financially healthy customers whose channel/category usage patterns lag their peers, at risk of disengaging further despite good financial standing.

CUSTOMER SEGMENTATION OVERVIEW

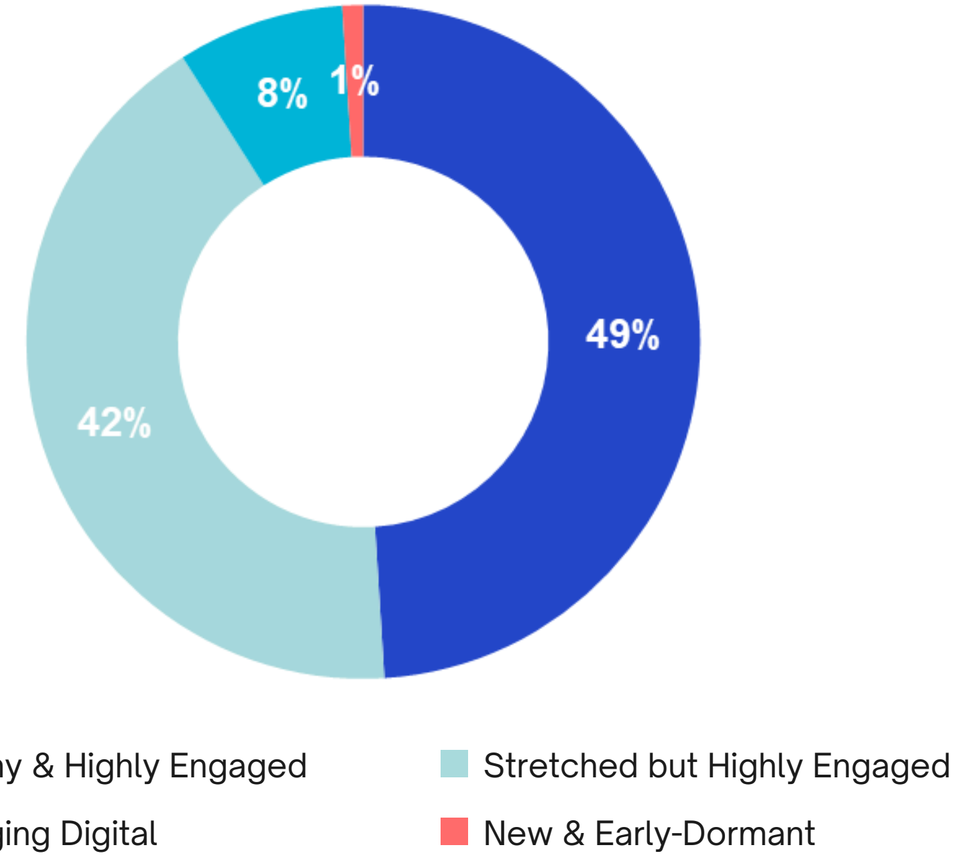
999
Total Customers

k = 4
Segments (silhouette-selected)

0.312
Best Silhouette Score

100%
Customer Coverage

Segment Share (999 Customers)



1 Feature Selection

Checked correlation/PCA across 9 candidate features; dropped 2 near-duplicates (category_diversity r=0.94 with engagement; spend_to_income_ratio r=-0.88 with health) to avoid double-counting one construct in the Euclidean distance.

2 Model Selection

Ran K-Means with k=3-8 on the remaining 7 features (standardized + median-imputed); selected k=4 for the highest silhouette score (0.312).

3 Coverage Validation

Confirmed all 999 customers are assigned to exactly one segment (100% coverage) before naming each segment from its own cluster statistics.

SEGMENT PROFILES

494 customers (49.4%)

Financially Healthy & Highly Engaged

69.4	78.0	0.01d	0.60
Health	Engagement	Recency	Spend/Income

The largest segment – financially secure and highly engaged; a strong candidate for wealth-building/investment product cross-sell.

416 customers (41.6%)

Financially Stretched but Highly Engaged

62.8	77.9	0.01d	0.81
Health	Engagement	Recency	Spend/Income

As engaged as the segment above but under greater spend-to-income pressure – contains 54 of the 59 system-wide 'Stressed & Highly Engaged' customers.

84 customers (8.4%)

Emerging Digital Customers

65.7	49.8	6.80d	0.61
Health	Engagement	Recency	Spend/Income

Few months of history, skews online – the lower engagement score reflects short tenure, not disengagement. Also holds the system's only 'Healthy & Disengaged' customer (Task 3).

5 customers (0.5%)

New & Early-Dormant Customers

65.5	38.7	24.4d	0.63
Health	Engagement	Recency	Spend/Income

Very new and largely inactive recently – should be monitored as an early churn-risk group.

RECOMMENDATION ACTIONS

6
Non-punitive Interventions

250
Max Reach (25.0%)

62
Min Reach (Product Suggestion)

59
'Danger Zone' Customers

Intervention	Defining Threshold	Reach (n / %)	Evidence Metric (independent driver)
Financial-planning Reminders	Health score ⩽ 25th percentile	250 (25.0%)	spend_to_income_ratio
Budgeting Tool	Spend/income ⩾ 75th percentile	250 (25.0%)	credit_utilization_ratio
Spend Alerts	Spending volatility ⩾ 75th percentile	250 (25.0%)	financial_health_score
Financial Education	Essential-spend ⩾75th pct AND spend/income ⩾ median	123 (12.3%)	credit_utilization_ratio
Digital-channel Nudges	Healthy customers, bottom 20% engagement	75 (7.5%)	online_spend_ratio
Product Suggestions	Health ⩾75th pct AND engagement ⩾75th pct	62 (6.2%)	total_spend_vnd

Methodology note: the first three rules use a 75th/25th-percentile threshold on a single variable ⇒ reach ≈25% by design, not coincidence. Each rule's driver strength (Cohen's d) is measured on an evidence metric DIFFERENT from its defining threshold (right column), to avoid circular self-validation.

VALIDATION & LIMITATIONS

K-Means segment vs. simple rule-based 2×2 (median-split health × engagement)

Segment	Healthy & Engaged	Healthy & Disengaged	Stretched & Engaged	Stretched & Disengaged
Healthy & Highly Engaged	230	195	44	25
Stretched but Highly Engaged	13	24	216	163
Emerging Digital	0	37	0	47
New & Early-Dormant	0	2	0	3

Darker color = more customers. K-Means recovers the rough structure of the 2×2 rule (two large groups splitting clearly along the engaged/disengaged columns), while also separating out a "new / short-tenure" group the 2×2 rule cannot see.

91 customers (9.1%)
Have <12 months of data (joined mid-year) – profiled on very few data points; treat as provisional.

81 customers (8.1%)
Missing a December record → recency was imputed with the median before standardizing/clustering.

Cross-check with Task 2: 'Stressed & Highly Engaged' (health<40, engagement⩾75) — 73 monthly records, 59 unique customers, computed independently on the same team-agreed thresholds — matches the 'Danger Zone' reported in Task 2.

Future Work

- → Month-by-month analysis (not just annual averages) to catch shifting trends.
- → Bootstrap resampling to test the stability of the K-Means segments.
- → Feed the 6 target groups into a next_month_low_health_flag prediction model (time-based train/test split).
- → Recompute percentile thresholds once real, non-synthetic data is available.